

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Total Marks: 25

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Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Introduction

Cloud computing helps organizations handle sudden increases in website traffic using **Auto-Scaling** and **Load Balancing**. These services ensure applications remain available even during peak demand.

Auto-Scaling

Auto-scaling automatically increases or decreases the number of servers based on demand.

- During seasonal sales, new servers are added automatically.
- When traffic decreases, extra servers are removed to reduce costs.

Load Balancing

A load balancer distributes incoming user requests evenly among multiple servers.

- Prevents any single server from becoming overloaded.
- Improves application speed and reliability.

Reducing Downtime

- If one server fails, traffic is redirected to healthy servers.
- Applications remain available without interruption.

Role of Monitoring Tools

Monitoring tools (such as Amazon CloudWatch or Azure Monitor):

- Track CPU, memory, and network usage.
- Trigger auto-scaling when resource usage exceeds thresholds.
- Send alerts for abnormal conditions.

Real-World Example

During **Amazon Great Indian Festival** or **Flipkart Big Billion Days**, millions of customers visit the website simultaneously. Auto-scaling launches additional servers, while load balancing distributes requests evenly, ensuring fast page loading and uninterrupted shopping.

Conclusion

Auto-scaling, load balancing, and monitoring together improve performance, reduce downtime, lower costs, and provide a better user experience.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Introduction

Cloud storage allows organizations to securely store backup data while ensuring durability, availability, and cost efficiency.

Durability and Data Redundancy

- Cloud providers store multiple copies of data in different locations.
- If one server or data center fails, another copy is available.
- This protects against hardware failures and disasters.

Encryption

- **Data at Rest:** Encrypts stored backup files.
- **Data in Transit:** Encrypts data while transferring over the internet.
- Prevents unauthorized users from reading sensitive information.

Access Control

- Role-Based Access Control (RBAC) gives permissions only to authorized users.
- Multi-Factor Authentication (MFA) adds extra security.

Cost Advantages

- No need to purchase expensive storage hardware.
- Pay only for storage used.
- Lower maintenance and operational costs.
- Easy scalability as storage needs increase.

Disaster Recovery Example

A company's primary data center is flooded. Backup data stored in the cloud is restored to another region, allowing business operations to continue with minimal downtime.

Conclusion

Cloud backup provides secure, reliable, and cost-effective data protection with strong disaster recovery capabilities.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction

A Content Delivery Network (CDN) improves application performance by delivering content from servers located close to users.

How CDN Improves Performance

- Stores copies of images, videos, and web pages on edge servers.
- Reduces the distance between users and data.
- Decreases loading time.

Edge Servers

- Deliver content from the nearest location.
- Reduce network congestion.
- Improve response time.

Global Availability

Cloud providers have data centers worldwide.

- If one region fails, another region continues serving users.
- Provides high availability and reliability.

Real-Time Streaming Example

Netflix stores video content on CDN edge servers worldwide. Users receive content from the nearest server, resulting in smooth video playback with minimal buffering.

Benefits

- Faster loading
- Lower latency
- Better user experience
- Reduced bandwidth usage

- High availability

Conclusion

CDNs and edge servers enable fast, reliable, and globally available applications by reducing latency and improving content delivery.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Financial institutions must protect sensitive customer information and comply with government regulations. Cloud providers offer built-in security and compliance services.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI-DSS
- GDPR
- SOC 2

Logging and Monitoring

- Records every user activity.
- Detects suspicious behavior.
- Supports auditing and investigations.

Encryption

- Encrypts customer data during storage and transmission.
- Protects confidential financial information.

Data Sovereignty

Organizations store data only in approved geographic regions to comply with local government regulations.

Real-World Example

A bank stores customer transaction records in an Indian cloud region. All data is encrypted, user activities are logged, and only authorized employees can access customer information, ensuring regulatory compliance.

Conclusion

Cloud compliance, encryption, monitoring, and proper data location help financial institutions securely manage sensitive financial information.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Containers package applications with all required libraries and dependencies, making deployment faster and more consistent.

Containers	Virtual Machines
Lightweight	Heavyweight
Share host OS	Separate OS for each VM
Fast startup	Slow startup
Lower resource usage	Higher resource usage

Kubernetes Orchestration

Kubernetes automates:

- Container deployment
- Scaling
- Load balancing
- Self-healing

- Rolling updates

Portability

Containers run consistently across:

- Public cloud
- Private cloud
- Hybrid cloud
- Local systems

Microservices Example

An **Amazon shopping application** uses separate containers for:

- User Login
- Product Catalog
- Shopping Cart
- Payment Service
- Order Management

Each service can be updated independently without affecting the others.

Conclusion

Containers and Kubernetes provide fast deployment, efficient resource utilization, scalability, portability, and simplified management, making them ideal for modern cloud-native applications.